3. Among the transition metals of 3d series, the one that has highest negative $\left(M^{2+}/M\right)$ standard electrode potential is

   a) Ti b) Cu c) Mn d) Zn

4. Which one of the following ions has the same number of unpaired electrons as present in V^{3+}?

   a) Ti^{3+} b) Fe^{3+}

   c) Ni^{2+} d) Cr^{3+}

5. The magnetic moment of Mn^{2+} ion is

   a) 5.92BM b) 2.80BM

   c) 8.95BM d) 3.90BM

6. The catalytic behaviour of transition metals and their compounds is ascribed mainly due to

   a) their magnetic behaviour

   b) their unfilled d orbitals

   c) their ability to adopt variable oxidation states

   d) their chemical reactivity

7. The correct order of increasing oxidizing power in the series

   a) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$ b) $Cr_2O_7^{2-} < VO_2^+ < MnO_4^-$

   c) $Cr_2O_7^{2-} < MnO_4^- < VO_2^+$ d) $MnO_4^- < Cr_2O_7^{2-} < VO_2^+$

8. In acid medium, potassium permanganate oxidizes oxalic acid to

   a) oxalate b) Carbon dioxide

   c) acetate d) acetic acid

9. Which of the following statements is not true?

   a) on passing H_2S, through acidified $K_2Cr_2O_7$ solution, a milky colour is observed.

   b) $Na_2Cr_2O_7$ is preferred over $K_2Cr_2O_7$ in volumetric analysis

   c) $K_2Cr_2O_7$ solution in acidic medium is orange in colour

   d) $K_2Cr_2O_7$ solution becomes yellow on increasing the P^H beyond 7

10. Permanganate ion changes to _____ in acidic medium

    a) MnO_4^{2-} b) Mn^{2+}

    c) Mn^{3+} d) MnO_2

XII U4-D-Block-Jerald Folder.indd 126 22-12-2022 19:49:38